

Luhai Tang

Worcester, MA | ltang@wpi.edu | (475) 287-3564 | www.linkedin.com/in/luhaitang | US Citizen

Education

Worcester Polytechnic Institute (WPI), Worcester, MA

B.S. Robotics Engineering, Manufacturing Engineering Minor, Electrical Power Systems Minor

May 2027

GPA: 4.0/4.0

Work/Experience

Advanced Manufacturing LLC | Automation and Manufacturing Intern - *Windsor, CT*

May 2025 - Jan 2026

- Designed a SOLIDWORKS replica of a THEO MA1-65 laser welder, replicating mass and load distribution within $\pm 5\%$, enabling safe motion testing and preventing potential damage to \$50k+ equipment.
- Developed and integrated an electrical I/O simulation system (LED indicators, emergency stop, laser pointer) with ABB DSQC1031 I/O expander, improving operation efficiency by 30% through real-time visual feedback.
- Programmed and optimized robotic welding routines in ABB RobotStudio (RAPID, AutoPath), achieving a 90% collision-free path success rate and reducing setup time by 40% compared to manual path teaching.
- Integrated 1 CNC and 1 AFSD machine, enhancing manufacturing automation capabilities and prototyping capacity.
- Collaborated on mechanical, electrical, and software integration across advanced manufacturing equipment, ensuring seamless interoperability between robotic and CNC systems.
- Developed custom control logic and programming for CNC, AFSD, and robotic systems, enabling safe, repeatable, and efficient operation in production testing.

Robotics Engineering Department | Robotics Lab Assistant - *Worcester, MA*

Sep 2024 - Present

- Repair and maintain more than 200 lab robots and other equipment, streamlining laboratory testing and improving reliability.
- Support 3 robotics lab sections by troubleshooting equipment and ensuring smooth operation of student experiments.
- Developed custom side projects and tools to streamline lab workflows, organization, and equipment management.

Projects

Autonomous Robot Navigation and Mapping System

Mar 2025 - May 2025

- Implemented SLAM with LiDAR in ROS, enabling autonomous environment mapping with $> 95\%$ positional accuracy.
- Designed and deployed an A* path planning algorithm to generate collision-free routes, reducing navigation time to target waypoints by $\sim 30\%$ versus baseline methods.
- Developed a pure pursuit controller that improved trajectory accuracy by 20% and enabled smooth, reliable robot motion.

UAV Regulation Intelligence Platform

Mar 2026 - May 2026

- Engineered a Flask-based regulatory knowledge platform with dynamic routes, modular Jinja templates, and region-specific JSON datasets to organize UAV certification, testing, and compliance information.
- Implemented keyword search and hierarchical browsing across over **100+ entries**, reducing manual lookup and time in use.
- Deployed the production site on Render with Gunicorn, environment-variable configuration, GitHub-hosted PDF assets, and admin-side content editing, supporting **50+ users** and enabling future development and full-scale implementation.

Skills

Programming Languages: Python, C, C++, MATLAB, ROS, Java, JavaScript, Linux, Git, RAPID

Robotic Software: ROS, ROS2, ABB Robot Studio

Mechanical and Electrical Design: SOLIDWORKS, CAD, FEA Analysis, KiCAD, LTSpice, MultiSim

Manufacturing: 3D Printing, CNC Programming

Hobbies: Music, Climbing, Volleyball, Video Creation, Traveling

Leadership & Activities

Pan Asian Association | External Vice President, Secretary, Performance Manager

Jul 2024 - Present

- Organizer of the Pan Asian Festival/Semi-Formal, achieving a 25% turnout increase with over 2,000 music festival attendees.
- Handle a budget of over \$100k+, coordinating with performance agencies, and handling \$20k+ contracts

Society of Asian Scientists and Engineers | Membership Chair, Events Committee

Sep 2023 - Present

- Developed SASE Roadmaps and organized events, improving member participation at Professional events by 15%
- Awarded 2025 Overall Strongest Chapter at the SASE STEM National Convention